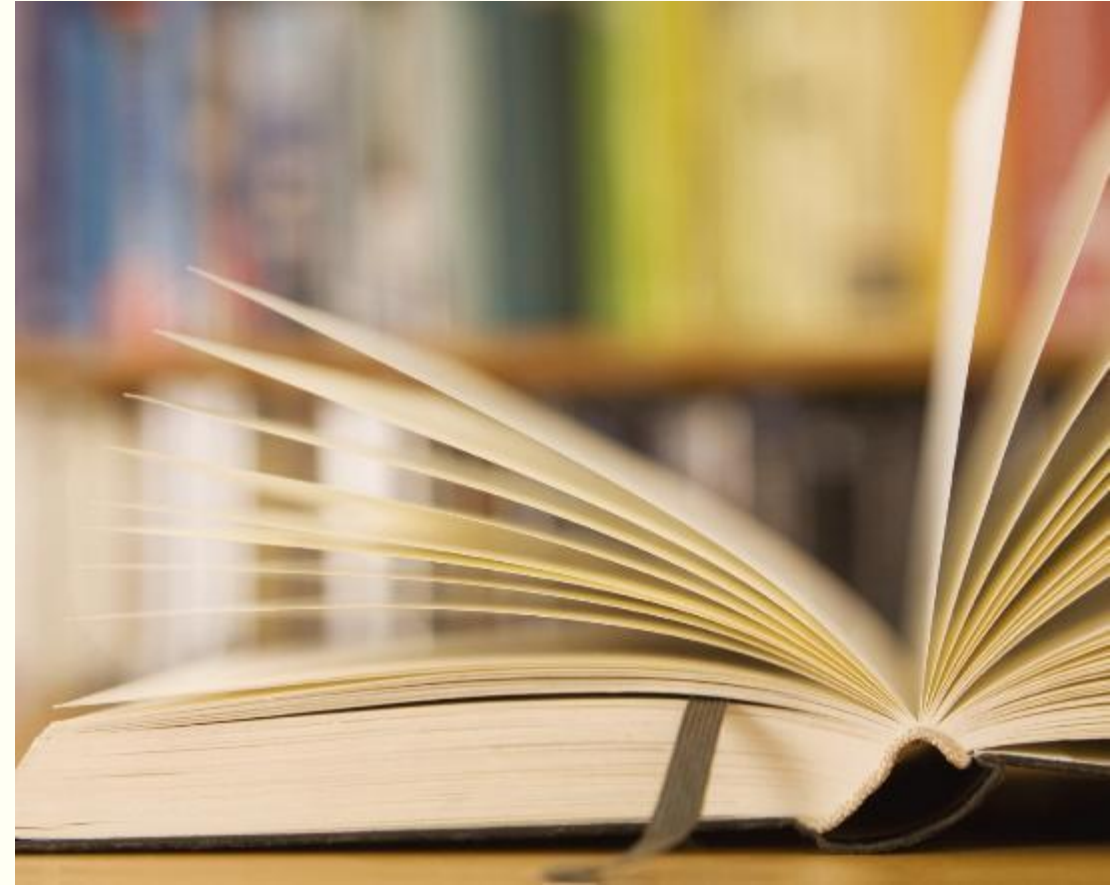




UNIVERSITY DROPOUT PREDICTION

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Introduction

- Early withdrawal affects student outcomes and institutional effectiveness
- Student retention remains a critical issue for universities (Tinto, 2012)
- Research shows these characteristics are associated with persistence and completion:
 - Demographic background
 - Academic preparation
 - Enrollment characteristics (Denley, 2019)



Overview of Data Used

- Project examines
 - Student dropout factors
 - Develops predictive models
 - Estimates likelihood that a student will withdraw
- Uses StudentInfo.csv dataset from The Open University Learning Analytics Dataset (OULAD), which includes these data features:
 - Demographic
 - Enrollment
 - Academic outcome (Kuzilek, 2017)



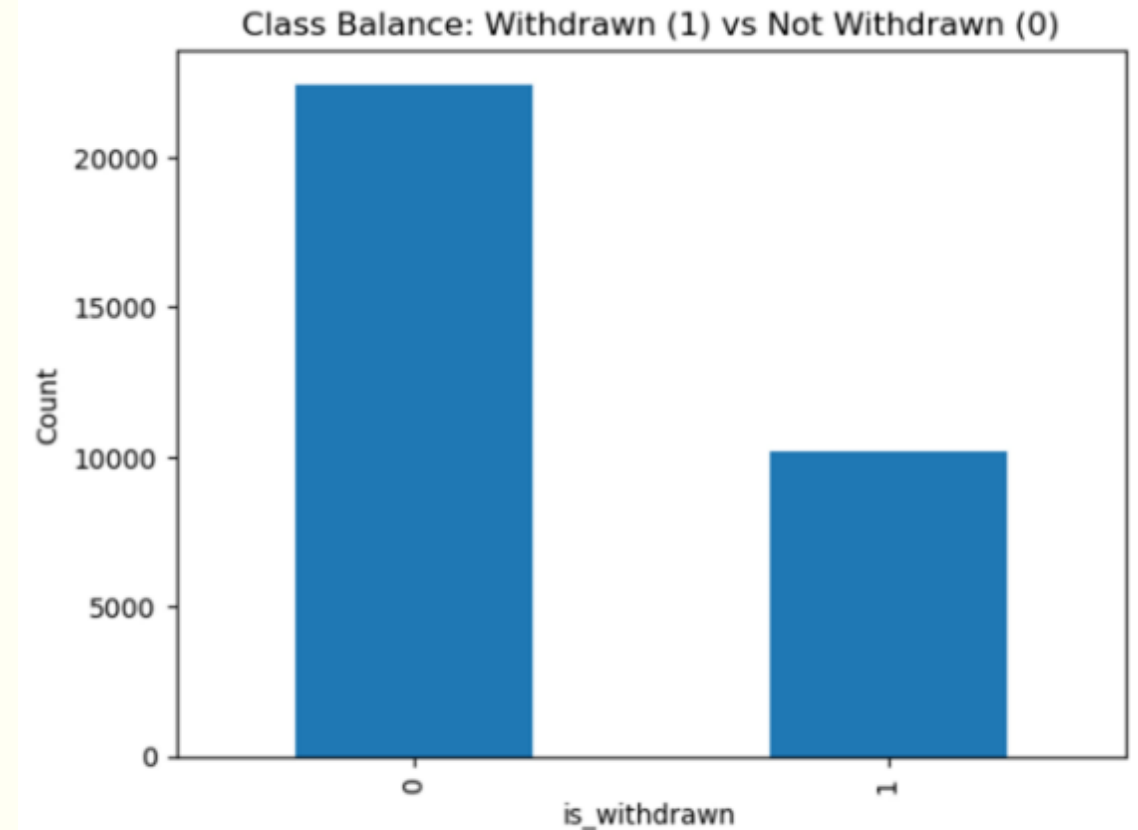
Methods of Analysis

- Exploratory Data Analysis (EDA)
 - Evaluate Data Quality
 - Review Class Balance
 - Identify Early Patterns – Predictors and Dropout Status (Han, 2012)
- EDA found a Class Imbalance existed with non-withdrawal outcomes occurring more frequently than withdrawals
 - Supports need to emphasize recall during model evaluation – failing to identify at-risk students carries practical consequences (James, 2021)
- Comparisons across demographic groups suggest these topics affect withdrawal rates
 - Age range of the student
 - Their highest education level
 - Disability status
 - Enrollment intensity (percentage of full-time coursework) (Tinto, 2012)



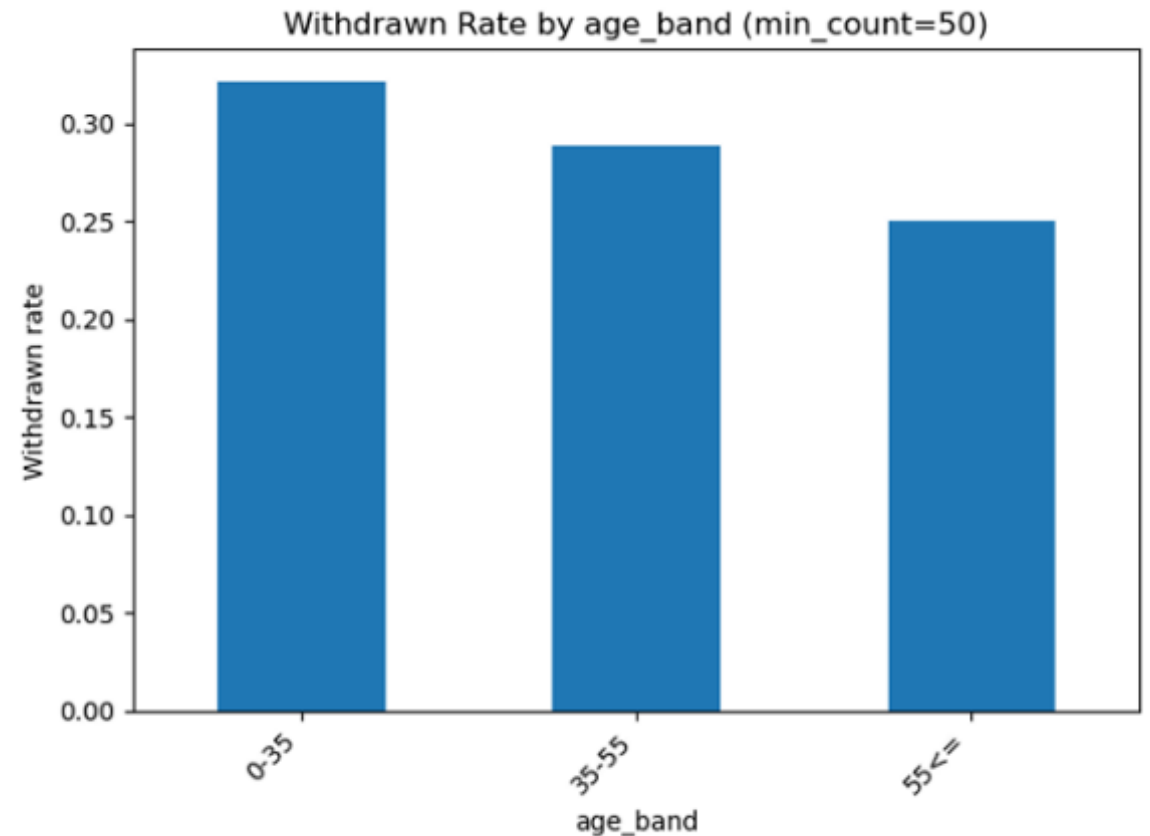
Exploratory Data Visualization

- Visualization: Distribution of Withdrawn vs Non-Withdrawn Students
- Purpose: Show class imbalance in the dataset
- Insight: Most students did not withdraw, which requires careful model evaluation
- Included chart: Bar chart of withdrawal distribution



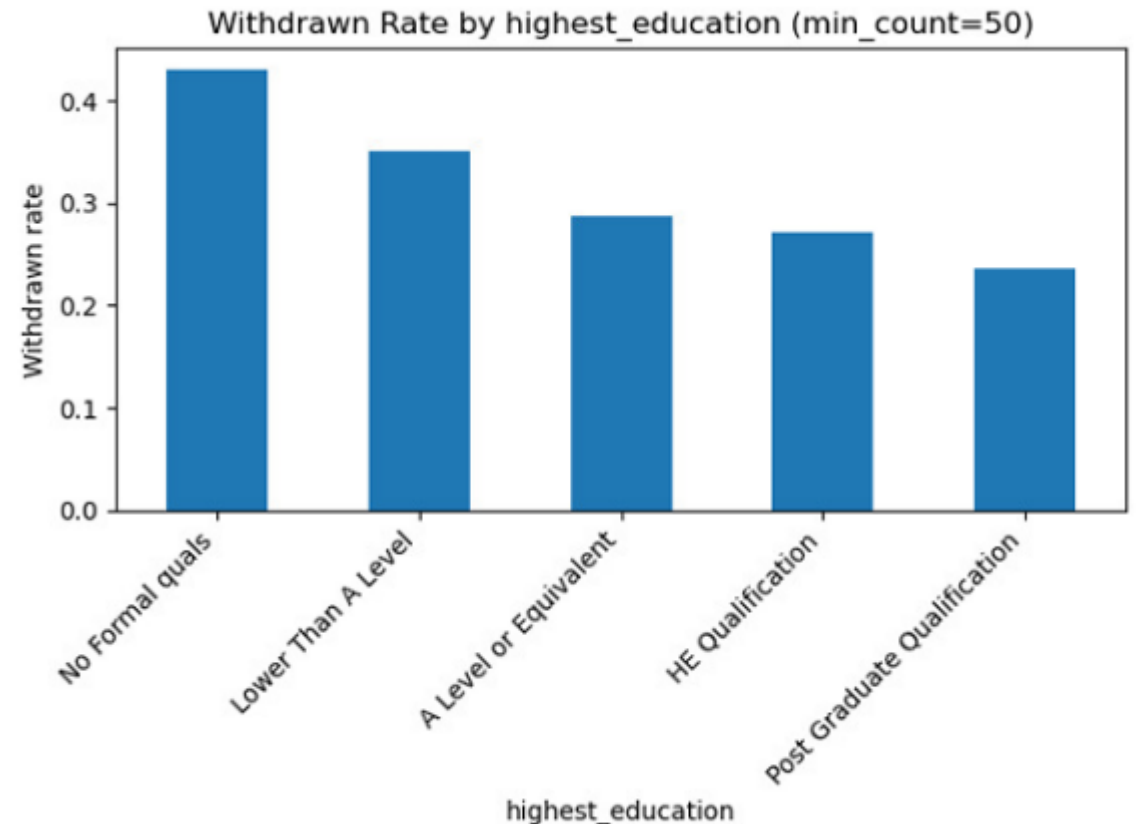
Withdrawal Rate by Age Band

- Visualization: Withdrawal rate across student age groups
- Purpose: Identify demographic patterns in student persistence
- Insight: Age differences may influence withdrawal likelihood
- Included chart: Stacked bar chart



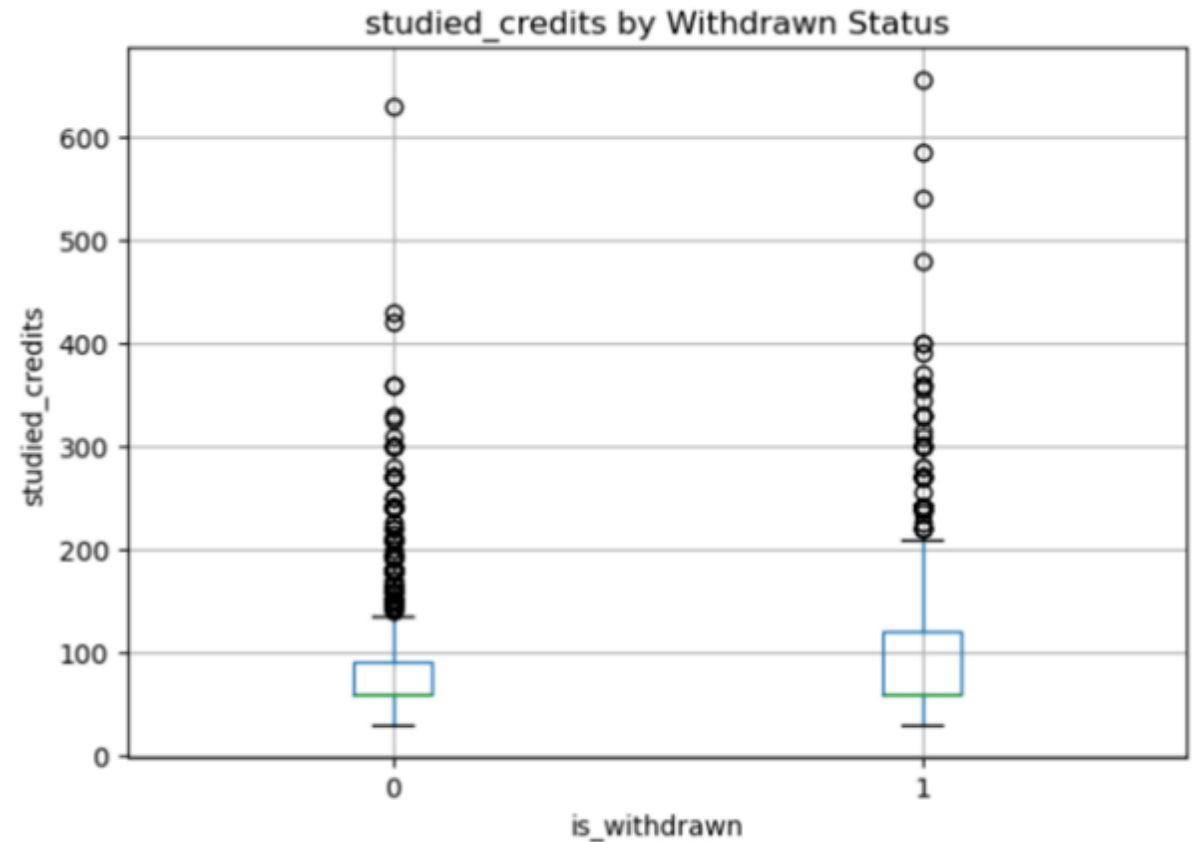
Withdrawal Rate by Education Level

- Visualization: Withdrawal probability by highest education level
- Purpose: Understand the relationship between academic preparation and dropout risk
- Insight: Students with lower academic preparation may have higher withdrawal risk
- Included chart: Bar chart



Distribution of Studied Credits

- Visualization: Distribution of studied credits by withdrawal status
- Purpose: Compare enrollment intensity across outcomes
- Insight: Students taking fewer credits may face higher dropout risk
- Included chart: Boxplot



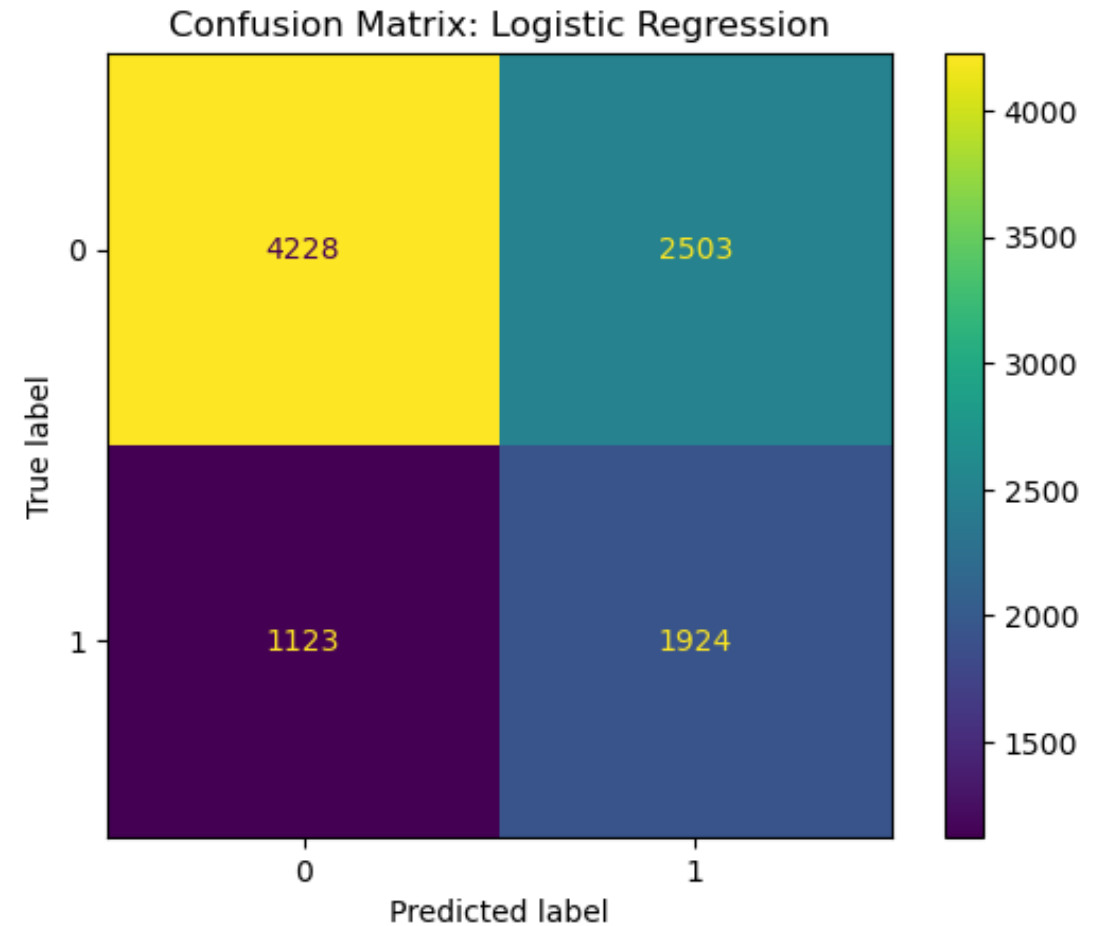
Model Performance Comparison

- Logistic Regression – Strong recall performance
- Random Forest – Highest accuracy but low recall
- XGBoost – Comparable performance to Logistic Regression
- Key takeaway: Logistic Regression best identifies at-risk students

	Model	Accuracy	Precision	Recall	F1	ROC_AUC
0	Logistic Regression	0.629	0.435	0.631	0.515	0.681
1	XGBoost	0.633	0.437	0.616	0.511	0.678
2	Random Forest	0.649	0.411	0.296	0.344	0.608

Confusion Matrix Interpretation

- Shows model predictions vs actual outcomes
- True positives represent correctly identified at-risk students
- False positives represent students flagged but not withdrawn
- False negatives represent missed at-risk students



Feature Importance Insights

- Course enrollment modules influenced withdrawal risk
- Academic preparation variables were strong predictors
- Disability status showed moderate predictive influence
- Socioeconomic indicators also affected persistence outcomes



Ethical Considerations

- Predictive models may reflect historical inequalities present in data
- Variables such as socioeconomic status or disability may correlate with withdrawal risk due to systemic factors
- Model predictions should support student success interventions rather than punitive decisions
- Human oversight and fairness monitoring are necessary for responsible implementation.



Conclusion

- Results suggest that student withdrawal risk is influenced by multiple related factors (Chawla, 2002):
 - Course enrollment patterns
 - Prior academic preparation
 - Disability status
 - Socioeconomic context
- Future improvements could include:
 - Additional feature engineering
 - Probability calibration
 - Cross-validation
 - Fairness monitoring
 - Incorporate behavioral engagement variables (including learning platform activity, assignment completion patterns)

References

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